Implementing Investment Strategies Under the Markowitz Framework

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Abstract

Our project focuses on the comparison of four different portfolios comprised of ten financial instruments each as well as a combined portfolio in order to determine which, under the Markowitz portfolio strategy, are most effective at maximizing returns under assumed acceptable levels of risk. We use our individual training data sets to compute a Markowitz strategy y over a range of risk parameters, plot our strategies against one another, and subsequently use validation data to evaluate the best risk parameter and testing data to assess strategy. Moreover, we analyze these results against the game theory framework wherein there is no notion of risk.

Our results find that financial currencies, as stable assets, yield less competitive returns for the same level of risk. Therefore, portfolios that were comprised heavily or solely of these instruments did not give higher payoffs as investment strategies. Using stocks alone as investment instruments notably increases risk, but compensates for this lessened stability with greater returns at every level. This model has the added value of being an investment strategy guide to both the novice or experienced investor. One can look at our model and note that we created vastly different portfolios comprised of varying financial instruments, as well as a combined portfolio, and observe how and why certain portfolios yield significantly higher reward for the same level of risk.

1 Introduction

The stock market has been around for well over 100 years. With an opportunity to increase one's wealth over a period of time by investing in companies' stocks the desire to maximize and minimize reward and risk, respectively, for each individual stock became a priority. The Markowitz model came about in 1952 to assist with stock portfolio optimization [2]. The Markowitz model can diversify and measure reward given assumed risk and thus inform the investor if an investment is a stable investment or too volatile to dive into. This can help prevent a dramatic loss in investment that could occur without the aid of an optimization model. This model will delve into the range of risk parameters,

and evaluate the most valuable stock portfolio for the investor based on the most optimal reward point for various levels of risk.

We have tested the recommended strategy that the Markowitz framework provided by comparing the behavior of four different portfolios with different investment strategies ranging from conservative to aggressive. Jimmy's portfolio is comprised only of foreign currencies, Emma's uses half foreign currencies and half stocks, Avery's contains ten randomly chosen stocks from the S&P 500, and Ryan's contains the top ten stocks of the Dow Jones. Our goal in choosing these combinations was to find the ideal balance of risk and reward, and to determine whether a portfolio with lots of stability and a lower, more dependable return would perform much better, much worse, or similarly to a portfolio with very risky yet lucrative elements.

2 Problem Formulation

We fix our vector μ and seek to optimize the reward via

$$\max_y \mu * \text{reward} - \text{risk}$$

We do this in such a way:

$$\max_y \mu\Big(\frac{1}{m}\sum_{i=1}^{m}\Big(\sum_{j=1}^{n}\widehat{a_{i_j}} * y_j\Big)\Big) - \frac{1}{m}\sum_{i=1}^{m}\Big|\sum_{j=1}^{n} a_{i_j} * y_j\Big|$$

where

$$\text{reward} = \frac{1}{m}\sum_{i=1}^{m}\Big(\sum_{j=1}^{n}\widehat{a_{i_j}} * y_j\Big)$$

$$\text{risk or } (b_i) = \sum_{j=1}^{n}\widehat{a_{i_j}} * y_j$$

(Note: risk is dependent on strategy) and where

$$\widehat{a_{i_j}} = a_{i_j} - f_j$$

and

$$f_j = \frac{1}{m}\sum_{i=1}^{m} a_{i_j}$$

$$\text{Strategy} = y * \mu$$

2.1 Assumptions

The Markowitz portfolio optimization model assumes a number of postulates:

1. Investors are rational and seek to maximize their utility
2. An Investor is risk averse
3. Risk of a portfolio is based on the variability of returns from said portfolio
4. An Investor either maximizes their portfolio return for a given level of risk or minimizes their risk for a given return
5. Analysis is based on a single period model of investment [2]

Additionally, we have limited the data of each instrument to 1000 points over the five-year time period 2006-2011 for the ease of splitting the data into the 70 percent training, 20 percent validation and 10 percent test subsets. If data values are missing for a specific date of a return on an investment, we will use the data from the preceding date to mitigate any MATLAB errors. We understand that both the way in which we have split our data and accounted for missing values will not present as accurate of a model, and we state below that, should we choose to improve upon or extend this model, we will change both of these assumptions.

2.2 Variable Definitions

Variables	**Definitions**
Currencies1	A variable (table) containing returns on Jimmy's portfolio (foreign currencies)
Emmas	A variable (table) containing returns on Emma's portfolio (half foreign currencies and half stocks)
Averys	A variable (table) containing returns on Avery's portfolio (ten randomly chosen stocks from the S&P 500)
Ryans	A variable (table) containing returns on Ryan's portfolio (top ten stocks of the Dow Jones)
Jimmy	An sxt table containing returns in Currencies1
Emma	An sxt table containing returns in Emmas
Avery	An sxt table containing returns in Averys
Ryan	An sxt table containing returns in Ryans
Total	An qxr table containing returns in all individual portfolios combined
s	The number of rows of the individual portfolio
t	The number of columns of the individual portfolio
q	The number of rows of the combined portfolio
r	The number of columns of the combined portfolio
Training1	The training data (70%) of Jimmy's portfolio
Training2	The training data (70%) of Emmas's portfolio
Training3	The training data (70%) of Avery's portfolio
Training4	The training data (70%) of Ryan's portfolio
Training	The training data (70%) of the combined portfolio
Validation1	The validation data (20%) of Jimmy's portfolio
Validation2	The validation data (20%) of Emmas's portfolio
Validation3	The validation data (20%) of Avery's portfolio
Validation4	The validation data (20%) of Ryan's portfolio
Validation	The validation data (20%) of the combined portfolio
Testing1	The testing data (10%) of Jimmy's portfolio
Testing2	The testing data (10%) of Emmas's portfolio
Testing3	The testing data (10%) of Avery's portfolio
Testing4	The testing data (10%) of Ryan's portfolio
Testing	The testing data (10%) of the combined portfolio
muv	A row vector that increments μ by .01 from 0 to 10
XX1	Keeps track of the decision vector (Jimmy) stored in it
XX2	Keeps track of the decision vector (Emma) stored in it
XX3	Keeps track of the decision vector (Avery) stored in it
XX4	Keeps track of the decision vector (Ryan) stored in it
XX	Keeps track of the decision vector (combined) stored in it

m	The number of rows of the individual training data
n	The number of columns of the individual training data
g	The number of rows of the combined training data
h	The number of columns of the combined training data
$f1$	The mean return on Training1
$f2$	The mean return on Training2
$f3$	The mean return on Training3
$f4$	The mean return on Training4
f	The mean return on Training
em	An mx1 row vector whose entries are 1's
en	An nx1 row vector whose entries are 1's
eh	An hx1 row vector whose entries are 1's
S1	Takes $f1$, repeats m times, and creates m identical rows consisting of the entries in $f1$
S2	Takes $f2$, repeats m times, and creates m identical rows consisting of the entries in $f2$
S3	Takes $f3$, repeats m times, and creates m identical rows consisting of the entries in $f3$
S4	Takes $f4$, repeats m times, and creates m identical rows consisting of the entries in $f4$
S	Takes f, repeats m times, and creates m identical rows consisting of the entries in f
AA1	The mean return matrix S1 subtracted from the matrix Training1
AA2	The mean return matrix S2 subtracted from the matrix Training2
AA3	The mean return matrix S3 subtracted from the matrix Training3
AA4	The mean return matrix S4 subtracted from the matrix Training4
AA	The mean return matrix S subtracted from the matrix Training
$I1$	An mxm identity matrix
zz-avery	The Game Theory payoff for Avery's testing data
i	Keeps track on iterations in the for-loop
$c1$	The cost vector for the matrix Training1
$c2$	The cost vector for the matrix Training2
$c3$	The cost vector for the matrix Training3
$c4$	The cost vector for the matrix Training4
c	The cost vector for the matrix Training
Ain1	The inequality constraint used for the matrix Training1
Ain2	The inequality constraint used for the matrix Training2
Ain3	The inequality constraint used for the matrix Training3
Ain4	The inequality constraint used for the matrix Training4
Ain	The inequality constraint used for the matrix Training

bin	The inequality constraint, which can be used for both individual and combined training matrices
Aeq1	The equality constraint, which can be used for all individual training matrices
Aeq	The equality constraint, which can be used for the combined training matrix (Training)
beq	The equality constraint, which can be used for both individual and combined training matrices
$d1$	The decision vector for matrix Training1
$d2$	The decision vector for matrix Training2
$d3$	The decision vector for matrix Training3
$d4$	The decision vector for matrix Training4
d	The decision vector for matrix Training
$y1$	The parts of the decision vector $d1$
$z1$	The parts of the decision vector $d1$
$y2$	The parts of the decision vector $d2$
$z2$	The parts of the decision vector $d2$
$y3$	The parts of the decision vector $d3$
$z3$	The parts of the decision vector $d3$
$y4$	The parts of the decision vector $d4$
$z4$	The parts of the decision vector $d4$
y	The parts of the decision vector d
z	The parts of the decision vector d
reward1(i)	Multiplies decision vector portion $y1$ by the column means of the return $f1$
reward2(i)	Multiplies decision vector portion $y2$ by the column means of the return $f2$
reward3(i)	Multiplies decision vector portion $y3$ by the column means of the return $f3$
reward4(i)	Multiplies decision vector portion $y4$ by the column means of the return $f4$
reward(i)	Multiplies decision vector portion y by the column means of the return f
risk1(i)	Takes the mean of the decision vector portion $z1$
risk2(i)	Takes the mean of the decision vector portion $z2$
risk3(i)	Takes the mean of the decision vector portion $z3$
risk4(i)	Takes the mean of the decision vector portion $z4$
risk(i)	Takes the mean of the decision vector portion z

myval(i)	Multiplies decision vector portion y by the column means of the return Validation
mytest(i)	Multiplies decision vector portion y by the column means of the return Testing
myval_avery(i)	Multiplies decision vector portion $y3$ by the column means of the return Validation3
mytest_avery(i)	Multiplies decision vector portion $y3$ by the column means of the return Testing
reward1	An array consisting of the values of reward1(i)
reward2	An array consisting of the values of reward2(i)
reward3	An array consisting of the values of reward3(i)
reward4	An array consisting of the values of reward4(i)
reward	An array consisting of the values of reward(i)
risk1	An array consisting of the values of risk1(i)
risk2	An array consisting of the values of risk2(i)
risk3	An array consisting of the values of risk3(i)
risk4	An array consisting of the values of risk4(i)
risk	An array consisting of the values of risk(i)
myval	An array consisting of the values of myval(i)
mytest	An array consisting of the values of mytest(i)
myval_avery	An array consisting of the values of myval_avery(i)
mytest_avery	An array consisting of the values of mytest_avery(i)

3 Model Development

We are utilizing a MATLAB-programmed simulation, specifically a linear program, to implement the Markowitz portfolio optimization model. We begin with a matrix of returns on various investments, where each row represents one time point (for our portfolios, returns are calculated daily) and each column represents a specific instrument (either a chosen stock or a foreign currency). Entries in this matrix represent the return on the instrument, calculated as a ratio of the close price over the open price in a given day. We then calculate another matrix of each instrument's mean returns using the repmat command, and subtract this from the original matrix to find deviation from the mean for each stock and currency. This helps us calculate a risk level for each instrument in our portfolio as a larger deviation represents a more volatile investment, which in turn yields more risk.

We define our risk parameter μ to take on incremental values from 0 to 10, increasing by .01 each time, and incorporate these values into a cost vector. $A_{\text{inequality}}$ represents the risk involved in each instrument and is derived from two standard basis vectors that represented the bounds of z_i, our decision. $B_{\text{inequality}}$ is set equal to 0 and z_i must be bound by the absolute value of

each instruments maximum deviations. A_{equality} and B_{equality} bind our strategy vector y so as to compute the required percentage of each instrument's return.

We test our individual portfolios separately in order to visualize the differences in our respective strategies. We begin by splitting each portfolio into 70% training data, 20% validation data, and 10% testing data, yielding 700 data points, 200 data points, and 100 data points respectively. We use the Markowitz portfolio code on the training data sets to ascertain our individual strategy vectors y for our defined μ. We find the best risk parameter using the rewards for each instrument and determining which data tuple of risk parameter and associated strategy yielded the largest return.

4 Results

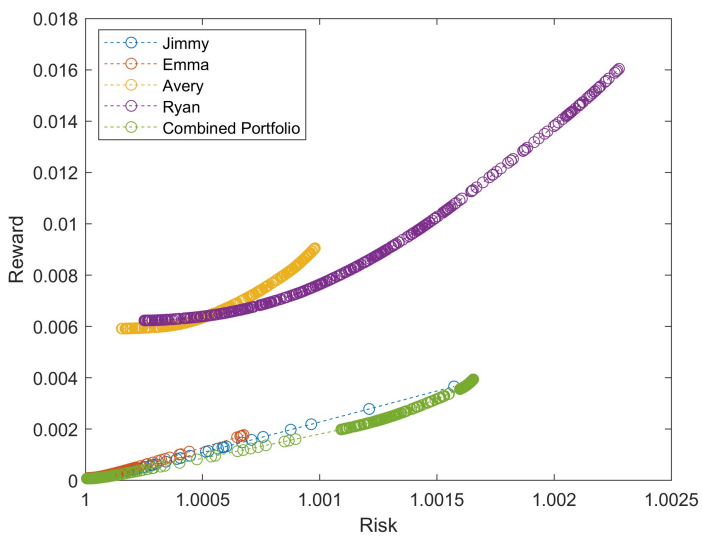

Figure 1: Plot of the Efficient Feasible Frontiers of Individual and Combined Portfolios

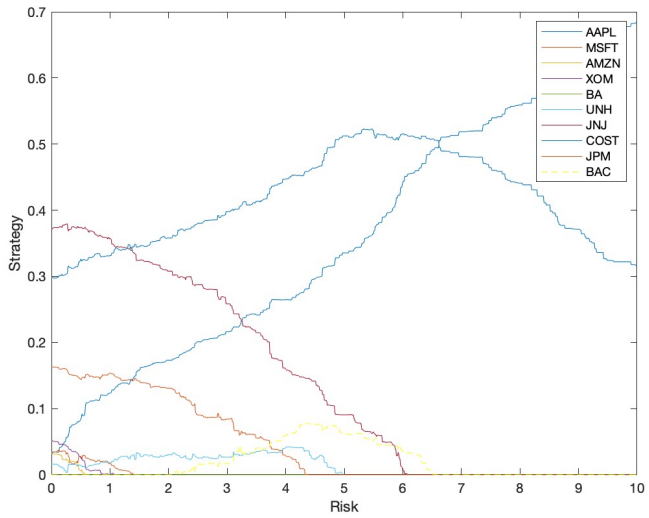

Figure 2: Plot of Risk vs. Strategy for Instruments in Avery's Portfolio

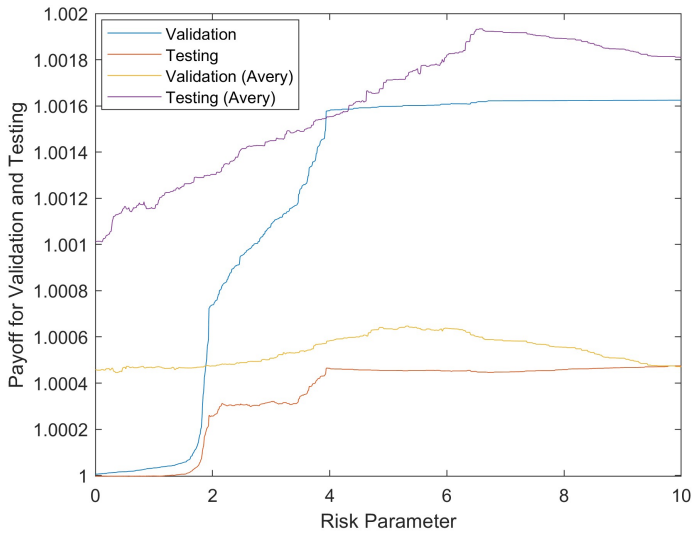

Figure 3: Plot of Combined Portfolio Validation and Avery's Validation Data Against μ

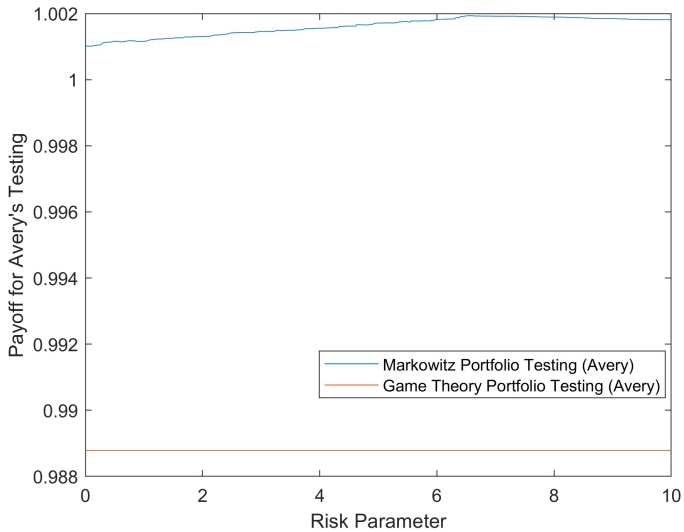

Figure 4: The Markowitz and Game Theory Portfolio Plot of Avery's Testing Data Against μ

Based on Figure 1, we see that Avery's portfolio had the smallest range of risk as well as high returns, which balanced all of our preferred strategies; Emma and Jimmy's portfolios were built around minimal risk and reliable returns while Avery and Ryan's attempted to maximize returns and accept whatever risk level was necessary. Since this portfolio did the best job of combining both priorities, we chose this as our preferred portfolio and used its validation and test data for comparison to the combined data, evaluation of the risk parameter, and evaluation of the strategy. Figure 2 details the strategy by illustrating which stocks are contained in the portfolio and their optimal proportions, accounting for risk levels.

In Figure 3, we see how the validation/testing data from the combined portfolio stacks up against the same data from Avery's portfolio. The testing data from Avery's portfolio implemented the strategy displayed in Figure 2. The graph clearly indicates that this strategy, when applied to the test set, yielded the highest returns for almost the entire range of risk parameters, meaning it was successful towards the goal of maximizing returns. The combined testing data produced significantly smaller returns – the lowest of all the data sets we tested. On the other hand, the validation data was used to nail down which risk parameter was the most effective. The combined portfolio showed a steep increase in return until reaching a risk parameter of about 4, where returns then abruptly stopped growing even as risk continued to increase. Avery's validation data reflected fairly consistent returns across all risk parameters but reached a peak at a slightly higher point somewhere around 6 (in later calculations of payoff, we determined the true best value to be mu equal to 6.6).

The final plot, Figure 4, compares the difference in payoff when implementing Markowitz portfolio theory versus game theory which does not evaluate risk at all. The game theory payoff for our preferred strategy came out to 0.988, which would not make an attractive investment as a payoff under 1 indicates that money is lost. However, the Markowitz payoff calculated with our ideal mu of 6.6 came out to 1.00193, representing a profit. This is both the more appealing and more realistic model as it allows the investor's money to grow and is a better replication of true market conditions, where risk is a hugely important factor to consider.

5 Robustness

Our model was fairly consistent throughout each trial for both individual and combined data sets in terms of the optimal risk parameter we found and the typical returns we saw amongst each investment strategy. Our model would likely tolerate changes in the data more accurately if we used a longer time period and a wider range of instruments. The years we pulled from did include some of the recession that took place in 2008, which likely affected the relative success of our two portfolios that included foreign currencies compared to those only made up of large-cap stocks. Repeating the same methods, using the same instruments, but taking data that spans a different 5 year period either pre- or post-recession would add some insight into the significance that our specific time period had on our conclusions. Doing even more trials this way and finding the averages for returns/risk and their standard deviations would make the model even more robust.

6 Model Evaluation

6.1 Model Strengths

Using a linear program to solve for our optimal portfolio offered a way to find the most efficient solution to our investment resources without the large amounts of trial and error that would be required without a program. The ability to add equality/inequality constraints ensured that with a reward of zero, risk would be as minimal as possible, and as the risk parameter grew reward would also grow proportionally to every tested value of mu. We found that the strategy it provided for appropriately allocating our resources was successful in providing significantly higher returns for the risk level we took on than those seen under the game theory framework, which is what we were hoping for as a reasonable investor will always require a higher return to compensate for increased risk.

Overall, the value this model brings to the table is predominantly its ability to compare several different risk parameters and determine the best reward an investor can expect at each of those points. This is useful because it is applicable to a wide range of investors, from those who prefer close to no risk and a lower, but guaranteed return, to those who are willing to accept even

the most volatile of investments with the potential for much higher long-term earnings. The way that we set up our individual portfolios to vary widely from each other in terms of risk/investment type added clarity to this takeaway as well - it made it easily visible which instruments were better or worse uses of our money. While this is a somewhat subjective way to classify the portfolios, we did clearly see that picking and choosing specific stocks of interest gave better returns than selecting the top 10 Dow stocks, and that using some foreign currencies for stability alongside some stocks to boost returns outperformed the portfolio currencies alone. Based on these comparisons, an investor could examine these four unique strategies and tune any one of them to their own preferred ratio of risk to reward.

6.2 Model Weaknesses

Our data collection could be improved upon by randomly pulling from the entire data set to create the training, validation, and test subsets. By pulling from the data in the way that we do, where the earliest 700 points make up training, the next 200 make up validation, and the most recent 100 make up the test set, we do not consider the entirety of the five year time period from which we sampled.

Another weakness of this model is that the portfolios were only looked at over a 5 year span. While 5 years might seem like a long enough time frame, general market evaluation is done over a larger period of time for improved accuracy in the case of adverse market health.

Additionally, a small number (less than ten) of data points were filled by the preceding date's return on the same instrument. We hypothesize that, due to the 2009 financial crisis, some data was not collected. This again does not create an entirely accurate portrayal of the Markowitz strategy y over our range of risk parameters for the time period.

7 Extensions of the Model

Given the opportunity to expand upon our current model, we would likely choose to evaluate returns over a larger time period and utilize more than 10 instruments per portfolio to further diversify them. This would help offset any abnormal highs or lows experienced in the market during the five years we pulled, which in the long run may be short-lived and insignificant relative to an investor's entire time in the market (say 30 years leading up to retirement). If our strategy for weighting instruments in our portfolios and determining our respective risk tolerances was based on a longer time period, it would be more reflective of a realistic, long-term investment plan.

A greater number of instruments would help add balance in a similar way; over time, different companies and industries will experience growth or losses at varying times, so diversifying a portfolio across many types of instruments would help to lower overall risk while still maintaining a consistent growth

level. While each of our portfolios is comprised of ten investments, notoriously successful portfolios have many more, such as the Dow Jones which contains 30 instruments and the S&P 500 with 500 instruments.

Another change we could implement to strengthen the model would be finding a way to include current, relevant market conditions in our calculations of risk levels. Our evaluation of each investment's risk is purely based on its own historical performance which is not always reflective of future behavior, and many risk measurement tools used in real-world portfolios are based on some other benchmark whose reliability is widely agreed upon. [1] Perceptions on things like emerging companies, inflation, foreign relations, etc. can have a large impact on market behavior, so it would be very useful if we had a way to incorporate these kinds of factors in our analysis.

Furthermore, implementing an ability to assess the effect of social media on stock market behavior would be beneficial. While we have focused on historical data, the ability for a CEO to send a tweet and influence the market day-to-day has to be taken into account when evaluating a stock portfolio both short term, and long term in today's market.

Moreover, another change we would like to implement would be the increase of more sustainable stocks that could outperform our own combined portfolio. With more data, and variation in these stocks we could more than likely find a better optimization of risk/reward for a portfolio that could compete with Emma's model. The ability to find another combination of stocks would certainly be an enhanced version of this model and give us more details into the true value of our stocks both singularly and combined that we could not perform with variations of our own portfolios.

8 Conclusions

Every person invests a little differently: an older woman, for example, may feel less inclined to assume a high level of risk compared to a young college graduate entering the financial market on a decent salary. Therefore, our evaluated best strategy is an evaluation purely based on which portfolio produced the highest reward for the lowest uniformly assumed risk.

We see that portfolios of financial currencies, as financial instruments, are low-risk, low-reward, stable assets, and gave less competitive returns, or lower reward for the same assumed risk, in contrast with Avery's or Ryan's portfolios (Figure 1). As our combined portfolio is comprised of roughly one-third of foreign currency assets, we did not outperform Avery's portfolio of diversified stocks or Ryan's portfolio of the top ten Dow Jones financial instruments. As Avery's portfolio produced the highest reward for the lowest risk, we utilized her strategy generated from the training data set to analyze differences in payoff.

Plotting the best risk parameter using the combined portfolio did not yield a higher payoff for the testing data, however Avery's portfolio testing data did yield a higher payoff. For a μ approximately equal to 6.6, the payoff of her

strategy is 1.00193. Taking Avery's portfolio testing data again and applying the game theory model that does not account for risk, we yield a payoff of 0.988 (Figure 4). We conclude that the Markowitz portfolio model yields the better payoff and considers the real notion of risk in investing.

Markowitz, while creating his portfolio theory, hypothesized that diversification should should aim to reduce the risk of a security, or financial instrument or asset, by limiting its variability and the covariance or interactive risk of two or more instruments in a portfolio. In our model, we clearly see that, while Emma's portfolio was a mix or diversification of stock options and financial currencies, the variability and covariance between any stock and the stable financial currencies yielded a lower performing strategy. Jimmy's portfolio did not have significant variability or interactive risk between his instruments, however, as his portfolio was comprised of solely financial currencies, it did not yield higher reward for the same level of risk as Ryan's or Avery's. We conclude that, for the average rational investor, one should diversify a portfolio via combinations of different instruments, but grouped in such a way to cap variability and covariance between assets. One option for someone following our model would be to have two portfolios: one that adheres to Avery's or Ryan's higher risk, higher reward stock option strategies and another that follows our combined strategy, comprised of financial currencies, yielding lower returns.

9 References

[1] J. Chen. *What are the 5 principal risk measures and how do they work?* 2022. URL: `https://www.investopedia.com/terms/r/riskmeasures.asp` (visited on 04/05/2023).

[2] Wikipedia. *Markowitz model.* URL: `https://en.wikipedia.org/wiki/Markowitz_model` (visited on 04/05/2023).

10 Source Code

10.1 Markowitz Portfolio Code

```
%DOWNLOAD and IMPORT .mat files to be able to run the data

%Source: https://finance.yahoo.com/

%The following are the table that consists of entries that represent the
%return on investments.
%The rows represent the daily date i (January 1, 2006 to January 1,
```

```
8  %2011), whereas the columns represent the investments
       j.
9
10 %"Jimmy" reads the (sxt) table in "Currencies1.xlsx"
       as the variable
11 %"Currencies1".
12 Jimmy = Currencies1;
13 [s t] = size(Jimmy);
14
15 %"Emma" reads the (sxt) table in "Emma's.xlsx" as the
       variable
16 %"Emmas".
17 Emma = Emmas;
18
19 %"Avery" reads the (sxt) table in "Avery's.xlsx" as
       the variable
20 %"Averys".
21 Avery = Averys;
22
23 %"Ryan" reads the (sxt) table in "Ryan's.xlsx" as the
       variable
24 %"Ryans".
25 Ryan = Ryans;
26
27 %"Total" combines "Currencies1", "Emmas", "Averys",
       and "Ryans into a (qxr)
28 %tabl
29 Total = [Currencies1 Emmas Averys Ryans];
30 [q r] = size(Total);
31
32 %"Training1" takes the first 700 rows of the different
        matrices,
33 %but still takes all the investments j.
34 %The first 700 rows denotes the 70 percent (training)
       of the total dataset.
35 Training1 = Jimmy(1:700,1:t);
36 Training2 = Emma(1:700,1:t);
37 Training3 = Avery(1:700,1:t);
38 Training4 = Ryan(1:700,1:t);
39 Training = Total(1:700,1:r);
40
41 %The following (line 42-46) converts training tables
       to training matrices.
42 Training1 = Training1{:,:};
43 Training2 = Training2{:,:};
44 Training3 = Training3{:,:};
```

```
45 Training4 = Training4{:,:};
46 Training = Training{:,:};
47
48 %The following (line 51-55) takes the next 200 rows of
       the different matrices,
49 %but still takes all the investments j.
50 %These 200 rows denote the 20 percent (validation) of
      the total dataset.
51 Validation1 = Jimmy(701:900,1:t);
52 Validation2 = Emma(701:900,1:t);
53 Validation3 = Avery(701:900,1:t);
54 Validation4 = Ryan(701:900,1:t);
55 Validation = Total(701:900,1:r);
56
57 %The following (line 58-62) converts validation tables
       to validation matrices.
58 Validation1 = Validation1{:,:};
59 Validation2 = Validation2{:,:};
60 Validation3 = Validation3{:,:};
61 Validation4 = Validation4{:,:};
62 Validation = Validation{:,:};
63
64
65 %The following (line 68-72) takes the last 100 rows of
       the different matrices,
66 %but still takes all the investments j.
67 %These 100 rows denote the 10 percent (testing) of the
       total dataset.
68 Testing1 = Jimmy(901:s,1:t);
69 Testing2 = Emma(901:s,1:t);
70 Testing3 = Avery(901:s,1:t);
71 Testing4 = Ryan(901:s,1:t);
72 Testing = Total(901:q,1:r);
73
74
75 %The following (line 76-80) converts testing tables to
       testing matrices.
76 Testing1 = Testing1{:,:};
77 Testing2 = Testing2{:,:};
78 Testing3 = Testing3{:,:};
79 Testing4 = Testing4{:,:};
80 Testing = Testing{:,:};
81
82 %muv is a row vector that increments mu by .01 from .0
       to 10.
83 muv = [.0:.01:10];
```

```

%Define XX's to be the decision vectors used for the plotting of
%the training matrices.
XX1 = [];
XX2 = [];
XX3 = [];
XX4 = [];
XX = [];

%The matrix Training1 has m rows and n columns.
[m n] = size(Training1);
%The matrix Training has g rows and h columns.
[g h] = size(Training);

%The following variables are the mean returns on investment on instruments
%in the training matrices.
%f's are row vectors of the size 1xm.
f1 = mean(Training1);
f2 = mean(Training2);
f3 = mean(Training3);
f4 = mean(Training4);
f = mean(Training);

%em, en, and eh are row vectors whose entries are 1's, which appear m, n,
%and h times respectively.
em = ones(1,m);
en = ones(1,n);
eh = ones(1,h);

%The following variables take the mean returns and repeat m times.
%This creates m same rows consisting of the entries in each f's.
%S's are of the size mxn.
S1 = repmat(f1,m,1);
S2 = repmat(f2,m,1);
S3 = repmat(f3,m,1);
S4 = repmat(f4,m,1);
S = repmat(f,m,1);

%The following variables aRE the mean return matrices S's subtracted from the training matrices with
%provided return values.
```

```
AA1 = Training1-S1;
AA2 = Training2-S2;
AA3 = Training3-S3;
AA4 = Training4-S4;
AA = Training-S;

%I1 is an mxm identity matrix.
I1 = eye(m);

%zz_avery is the Game Theory payoff for Avery's testing data.
%To check, run the gametheoryp2.m MATLAB file.
zz_avery = 0.9888;


%The following for-loop is intended for keeping track of the reward and risk coordinates (for plotting).

%Note that size(muv,2) returns the number of columns in muv(which is 1001),
%because 2 refers to the column dimension of muv in this case.
for i = 1:size(muv,2)
    %shows the i-th iteration.
    i

    %The following is the calculation for the reward and risk coordinates
    %using the training matrices.

    %c's is the cost vector for the matrix A.
    c1 = [muv(i)*f1 em*(-1/m)]';
    c2 = [muv(i)*f2 em*(-1/m)]';
    c3 = [muv(i)*f3 em*(-1/m)]';
    c4 = [muv(i)*f4 em*(-1/m)]';
    c = [muv(i)*f em*(-1/m)]';

    %Ain's and bin are provided and used for the inequality constraints
    %(for the training matrices).
    %Ain(1-4) is of the size (2m)x(m+n), Ain is of the size
    %(2m)x(m+h), bin is of the size (2m)x1.
    Ain1 = [-AA1 -I1; AA1 -I1];
    Ain2 = [-AA2 -I1; AA2 -I1];
    Ain3 = [-AA3 -I1; AA3 -I1];
```

```
    Ain4 = [-AA4 -I1; AA4 -I1];
    Ain = [-AA -I1; AA -I1];
    bin = [zeros(2*m,1)];

    %Aeq, Aeq1, and beq are provided and used for the equality constraint
    %(for the training matrices).
    %Aeq1 is of the size 1x(m+n), Aeq is of the size 1x(m+h), and beq is a
    %scalar.
    Aeq1 = [en 0*em];
    Aeq = [eh 0*em];
    beq = 1;

    %The following linprog commands are provided, using the
    %inequality and the equality constraints.
    [d1 val1] = linprog(-c1, Ain1, bin, Aeq1, beq, zeros(m+n,1))
    [d2 val2] = linprog(-c2, Ain2, bin, Aeq1, beq, zeros(m+n,1))
    [d3 val3] = linprog(-c3, Ain3, bin, Aeq1, beq, zeros(m+n,1))
    [d4 val4] = linprog(-c4, Ain4, bin, Aeq1, beq, zeros(m+n,1))
    [d val] = linprog(-c, Ain, bin, Aeq, beq, zeros(m+h,1))

    %y(1-4) are the decision vector portion of the size nx1, whereas y is of the
    %size hx1.
    y1 = d1(1:n);
    y2 = d2(1:n);
    y3 = d3(1:n);
    y4 = d4(1:n);
    y = d(1:h);

    %reward(i)'s is the multiplication of the decision vector y' by the
    %column mean return vector f'.
    %reward(i)'s are scalars.
    reward1(i) = y1'*f1';
    reward2(i) = y2'*f2';
    reward3(i) = y3'*f3';
    reward4(i) = y4'*f4';
    reward(i) = y'*f';
```

```
    %z's are the decision vector portion of the size mx1.
    z1 = d1(n+1:m+n);
    z2 = d2(n+1:m+n);
    z3 = d3(n+1:m+n);
    z4 = d4(n+1:m+n);
    z = d(h+1:m+h);

    %risk(i)'s are the mean of the entries in the vector z's.
    %risk(i)'s are scalars.
    risk1(i) = mean(z1');
    risk2(i) = mean(z2');
    risk3(i) = mean(z3');
    risk4(i) = mean(z4');
    risk(i) = mean(z');

    %XX's keep track of the decision vectors stored in it.
    XX1 = [XX1 d1];
    XX2 = [XX2 d2];
    XX3 = [XX3 d3];
    XX4 = [XX4 d4];
    XX = [XX d];

    %myval(i) is the payoff/reward for the combined validation data.
    myval(i) = y' *mean(Validation)';

    %mytest(i) is the payoff/reward for the combined testing data.
    mytest(i) = y' *mean(Testing)';

    %myval_avery(i) is the payoff/reward for Avery's validation data.
    myval_avery(i) = y3' *mean(Validation3)';

    %mytest_avery(i) is the payoff/reward for Avery's testing data.
    mytest_avery(i) = y3' *mean(Testing3)';

    %gt_avery(i) is the payoff/reward for Avery's testing data (Game Theory
    %version).
    gt_avery(i) = zz_avery;
```

```
end


%Figure 1
%The following 5 lines create a labelled plot of the Efficient Frontier
%graph (or Reward vs Risk graph) of the four individual portfolios and the
%combined portfolio.
figure('Name','The Efficient Frontier for the provided data set')
plot(reward1,risk1,'--o',reward2,risk2,'--o',reward3,risk3,'--o',reward4,risk4,'--o',reward,risk,'--o')
legend('Jimmy','Emma','Avery','Ryan','Combined Portfolio')
xlabel('Risk')
ylabel('Reward')

%Figure 2
%The following 5 lines create a labelled plot of Payoff for Validation and
%Testing vs Risk Parameter of the combined portfolio and Avery's portfolio.
figure('Name','Risk paramater vs reward')
plot(muv,myval,'-',muv,mytest,'-',muv,myval_avery,'-',muv,mytest_avery,'-')
legend('Validation','Testing','Validation (Avery)','Testing (Avery)')
xlabel("Risk Parameter")
ylabel("Payoff for Validation and Testing")

%Figure 3
%The following 5 lines create a labelled plot of Payoff for Avery's Testing
%vs Risk Parameter (Markowitz Portfolio and Game Theory Portfolio).
figure('Name','Risk paramater vs reward')
plot(muv,mytest_avery,'-',muv,gt_avery,'-')
legend('Markowitz Portfolio Testing (Avery)','Game Theory Portfolio Testing (Avery)')
xlabel("Risk Parameter")
ylabel("Payoff for Avery's Testing")
```

10.2 Game Theory Portfolio

```
%DOWNLOAD and IMPORT .mat files to be able to run the data

%Source: https://finance.yahoo.com/

%The following are the table that consists of entries that represent the
%return on investments.
%The rows represent the daily date i (January 1, 2006 to January 1,
%2011), whereas the columns represent the investments j.

%"Avery" reads the (sxt) table in "Avery's.xlsx" as the variable
%"Averys".
Avery = Averys;
[s t] = size(Avery);

%A takes the last 100 rows of the different matrices, but still takes all
%the investments j.
%These 100 rows denote the 10 percent (testing) of the total dataset.
A = Avery(901:s,1:t);
%The following converts Avery's testing table to testing matrix.
A = A{:,:};

%The matrix A has m rows and n columns.
[m n] = size(A);

%em and en are row vectors whose entries are 1's, which appear m and n
%times respectively.
em = ones(m,1);
en = ones(n,1);

%c is the cost vector for the matrix A.
c = [0*en' 1]';
%Ain and bin are provided and used for the inequality constraints (for the matrix A).
Aineq = [-A em];
bineq = 0*em;
```

```
%Aeq and beq are provided and used for the equality constraints (for the matrix A).
Aeq = [en' 0];
beq = 1;
%We need to take into account that the decision variables are bounded below by zero but the value of the games is not (it is free).
lb=[0*en' -Inf]';

%linprog command (for the matrix A) is provided, considering the inequality and the equality constraints.
[xx zz_avery] = linprog(-c, Aineq, bineq, Aeq, beq, lb);

%The following displays the strategy for the column player.
display('Strategy for Column Player')
xx

%The following displays the payoff.
display('Payoff for either player')
%Be sure to negate the result since we are solving max zz_avery, which is the same as -min-zz_avery.
-zz_avery
```